A FINITE BIRATIONAL MORPHISM OF INTEGRAL SCHEMES WITH NORMAL TARGET IS AN ISOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The canonical Stacks tag of the counted statement is 0AB1 (morphisms-lemma-finite-birational-over-normal). Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the tag is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The statement is self-contained: integral scheme, normal scheme, finite and integral morphism and birational morphism are all standard and none of them is introduced by an immediately preceding definition of this chapter that the statement quotes, so no context span is attached.

Editorial note. The parenthetical (or even integral) in the statement is the author's own remark that finiteness is used only through integrality; it is kept verbatim and the proof indeed uses integrality alone.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of sources/stacks-chow/preamble.tex lines 14-18 and 154-217. No macro was invented or redefined.

Allowed standard prerequisites (editorial).

- Integral closure and the definition of a normal scheme. Algebra, Section labelled algebra-section-normal (canonical Stacks tag 00G3); used at proof lines 14640-14641. Supplies that for a normal scheme the ring of sections O_Y(V) of an affine open is integrally closed in its fraction field, which is the step that forces O_X(U) = O_Y(V).
- A finite morphism is integral, and integrality is checked on an affine open cover. Morphisms, Lemma labelled morphisms-lemma-integral-locally-finite and the definition of an integral morphism in the same chapter; used at proof lines 14641-14643. Gives that every element of O_X(U) is integral over O_Y(V), i.e. lies in the integral closure of O_Y(V).
- A birational morphism of integral schemes is an isomorphism over a dense open, so the induced map on function fields is an isomorphism. Morphisms, Section labelled morphisms-section-birational (canonical Stacks tag 01RJ); used at proof line 14639. Supplies that O_Y(V) -> O_X(U) is injective and induces an isomorphism of fraction fields, so both rings sit inside one common field.

Lemma 0.1. *A finite (or even integral) birational morphism $f : X \to Y$ of integral schemes with Y normal is an isomorphism.*

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Proof. Let $V \subset Y$ be an affine open with inverse image $U \subset X$ which is an affine open too. Since f is a birational morphism of integral schemes, the homomorphism $\mathcal{O}_Y(V) \to \mathcal{O}_X(U)$ is an injective map of domains which induces an isomorphism of fraction fields. As Y is normal, the ring $\mathcal{O}_Y(V)$ is integrally closed in the fraction field. Since f is finite (or integral) every element of $\mathcal{O}_X(U)$ is integral over $\mathcal{O}_Y(V)$. We conclude that $\mathcal{O}_Y(V) = \mathcal{O}_X(U)$. This proves that f is an isomorphism as desired. $\square$